

NCBA LOOP CASE STUDY

Country Integration Service

Integrations and Microservices Engineer

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1. Spring Boot application setup

Framework and dependencies

Java 25, Spring Boot 4.1.1, Maven wrapper; Web, validation, JPA, MySQL, Actuator and Flyway.

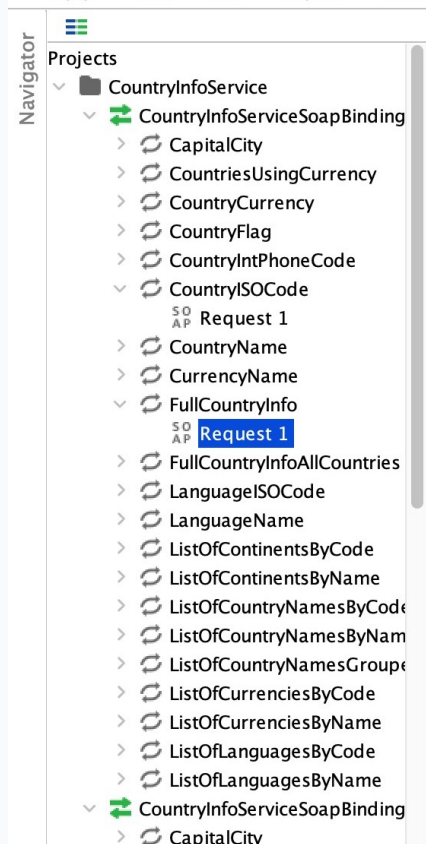
Separation of concerns

Controller handles HTTP; service coordinates operations; SOAP client/transport own integration; repository accesses MySQL.

Configuration and startup

Environment variables supply credentials and endpoint. Flyway creates fresh schemas; Hibernate validates them.

2. SoapUI import, request and assertions



CountryISOCode • actual Tanzania exchange

Request

```
<web:sCountryName>Tanzania</web:sCountryName>
```

HTTP 200 response

```
<m:CountryISOCodeResult>TZ</m:CountryISOCodeResult>
```

Passed: SOAP response • no fault • XPath result TZ

3 cases / 9 assertions: ISO Tanzania, FullCountryInfo TZ, ISO South Africa.

Reproduce: ./scripts/test-soapui.sh

3. Receive JSON and normalize the name

REST boundary

POST /api/countries accepts {"name":"kenya"}. Bean Validation rejects blank names and malformed JSON with 400.

Required first lookup

Trim and collapse whitespace, then sentence case: kenya → Kenya; south africa → South africa.

HTTP response

Successful import returns 201, the persisted country/languages and a Location header. Unknown country returns 404.

4. Resolve ISO with provider-compatible names

CountryISOCode request

Send sCountryName in the SOAP envelope. Parse CountryISOCodeResult and validate the ISO code.

Correctness fix

After an unknown sentence-case result, title case includes punctuation: Guinea-Bissau and Papua-New Guinea now resolve.

Canonical spelling and cache

If needed, use the provider name catalogue with a six-hour TTL. Moldova, Republic of resolves to MD. Transport errors never trigger name-format fallback.

5. Fetch and persist full country information

FullCountryInfo request

Use the resolved ISO code as sCountryISOCode. Read country fields and all returned languages.

Defensive parsing

Namespace-aware XML parsing, escaped request values, prohibited DOCTYPE/external entities, SOAP-fault and malformed-result checks.

Database consistency

SOAP finishes before save. A unique ISO constraint handles concurrent duplicates; the service maps the losing save to 409.

6. JPA entities and request DTOs

CountryInfo and Language are JPA entities

CountryInfo persists country identity/details. Language persists ISO/name and its country relationship.

DTOs describe incoming requests

CreateCountryRequest and UpdateCountryRequest validate the REST input. They are distinct from database entities.

Persistence behavior

Generated IDs, unique country ISO, cascading save/delete and orphan removal. Tests verify language replacement and eager read paths.

7. CRUD endpoints and bounded reads

Method	Route / behavior	Success
POST	/api/countries; import and save	201 + Location
GET	/api/countries?page=0&size=100	200 JSON array
GET	/api/countries/{id}	200 country
PUT	/api/countries/{id}; replace details/languages	200 updated
DELETE	/api/countries/{id}; cascade languages	204 empty

Zero-based pages; size 1–100. Read until an empty page for all countries. Invalid bounds return 400. ISO and country ID stay immutable on update.

8. Kubernetes deployment scripts

Source-built image

Multi-stage Java 25 build; non-root runtime. A tag derived from Docker build inputs prevents stale local images.

Deploy and scale

`./scripts/deploy.sh` creates namespace/Secret, waits for MySQL, runs migrations with one app replica, then scales to two.

Cluster portability

Docker Desktop uses the local image; kind/minikube import it. Remote contexts require `APP_IMAGE` with registry path and `PUSH_IMAGE=true`.

9. Run, test and deploy

From the project root

- 1 `cp .env.example .env` — replace both password placeholders
- 2 `./mvnw clean verify` — 55 tests; no Docker or external DB
- 3 `docker compose up -d --build` — API localhost:8081
- 4 `./scripts/deploy.sh` — Kubernetes, two replicas
- 5 Port-forward Service 8082:8080; run `./scripts/verify.sh`

All 64 tests

`./mvnw -Pmysql-it clean verify` — Docker required for temporary MySQL.

Existing pre-Flyway database: backup first, explicit baseline once, then return baseline flag to false.

10. Troubleshoot Kubernetes

Image and startup

Check pod events, image tag and rollout; load the image for your cluster. Inspect logs for schema validation and datasource failures.

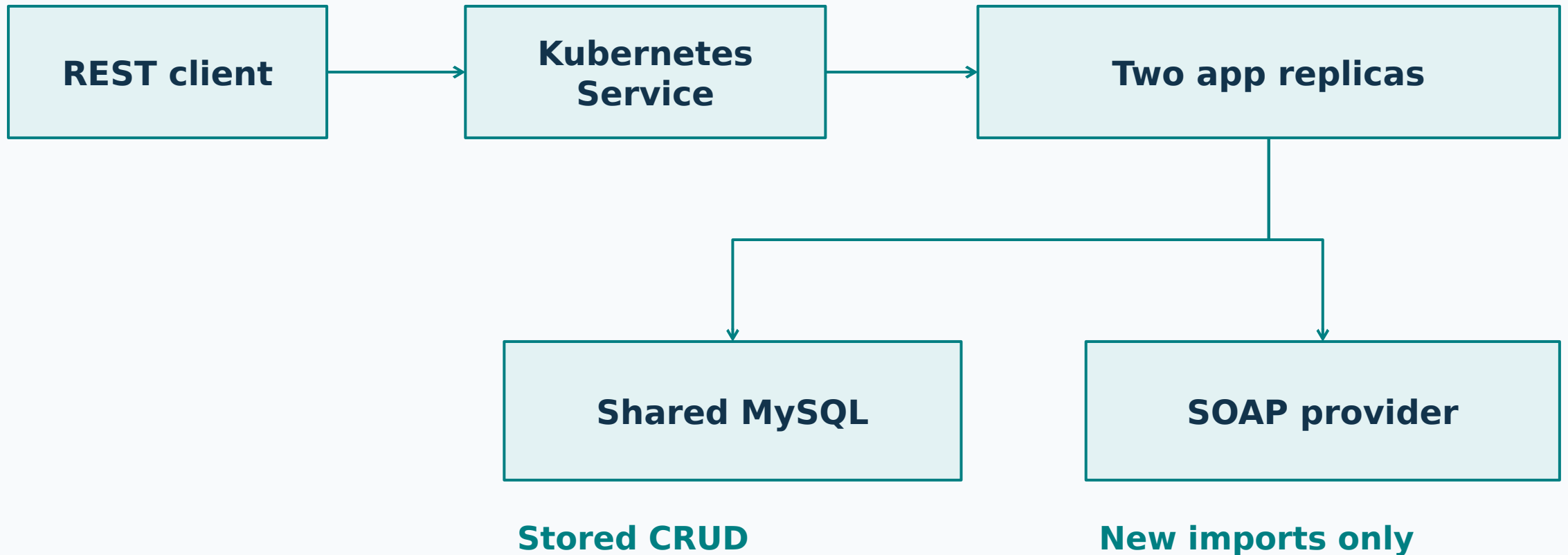
Readiness and upstream failures

Readiness includes MySQL; liveness does not. Check PVC, database access, SOAP timeouts, circuit state and outbound network access.

Resources and diagnostics

Inspect limits/probe timeouts, restart counts and Service endpoints. Use request IDs, business outcomes, Prometheus targets and Grafana.

System design and integration trade-offs



Synchronous import gives an immediate result. Bulk queuing is a proposed extension requiring durable jobs, idempotency and dead-letter handling.

Scaling: measured local read workload

Measurement	Observed local result
Requests / concurrency	200 reads / 8 workers; all HTTP 200
Service distribution	101 requests to one pod; 99 to the other
Rate / latency	9.471 requests/s; p50 450 ms; p95 3,106 ms
Test scope	21.118 s sample on one Docker Desktop node

Stateless replicas and shared MySQL support horizontal scaling. SQL pagination bounds reads; provider-name caching avoids repeated catalogue lookups.

HPA 2-4 replicas is configured, but this cluster lacks the metrics API. Autoscaling, sustained capacity and multi-node failover remain unverified.

Resilience and HTTP error handling

Bounded external calls

5 s connection timeout; each request up to 8 s. At most two attempts, 300 ms apart. Overall SOAP sequence budget: 20 s.

Circuit breaker and degraded operation

10-call window; minimum five, 50% failure threshold; open 30 s, two half-open calls. Stored CRUD remains available during SOAP outages.

Explicit HTTP outcomes

400 invalid input, 404 missing country, 405 method, 406 Accept, 409 duplicate, 415 media type, 502 invalid upstream, 503 unavailable, 504 timeout.

Logs, metrics and deployed monitoring

Structured troubleshooting logs

Request ID, method/path/status/duration; import/ISO/save outcomes, duplicate detection and failure status. Update/delete log flushed results. Credentials and bodies excluded.

Business and HTTP metrics

country.operations outcomes, country.import.duration, HTTP request rate/latency and pod identity. Actuator exposes health and Prometheus metrics.

Verified local monitoring

Prometheus scraped two healthy targets and loaded availability/5xx rules. Grafana provisioned the country-service dashboard. Storage is ephemeral; local Viewer access only.

64 automated tests passed

Suite	Tests	Evidence
Service / failure	21	Names, duplicate races, errors and CRUD
SOAP client	10	Canonical cache, namespaces, faults, XML safety
Controller / transport	18	POST/PUT, validation, timeout/budget/circuit
Context / H2 persistence	6	Startup and JPA behavior
Actual MySQL	9	Persistence, concurrency and Flyway migration

Default: ./mvnw clean verify → 55 tests, no Docker required

Full: ./mvnw -Pmysql-it clean verify → 64 tests, Docker required

Final deployment verification

Data preserved and schema validated

Existing Kenya, Tanzania and South Africa survived Flyway adoption. Baseline was explicitly enabled once and then disabled. A private dump restored into disposable MySQL with all six countries and languages.

Correctness and shared persistence

Guinea-Bissau → GW, Papua-New Guinea → PG, Moldova → MD: each created with 201 and read through the other replica. Duplicate 409; unknown 404.

Ready application and evidence

Two ready backends; health/readiness/liveness UP; zero app/MySQL restarts. Pagination, business metrics and internal Service distribution verified.

Submission and production boundaries

Assessment deliverables

Source, tests and CI; SoapUI project/results; Docker and Kubernetes scripts; deployment/troubleshooting guides; requirement mapping; presentation and PDF.

Proposed production architecture

Multiple nodes/zones and a managed HA database; authenticated TLS ingress; managed secrets; durable monitoring and alert destinations; tested backups and recovery.

Remaining validation

Local results do not prove HA or peak capacity. HPA behavior, sustained mixed load/fault tests, production recovery exercises and upstream HTTPS need the corresponding environment.

github.com/kimtour/country-integration-service